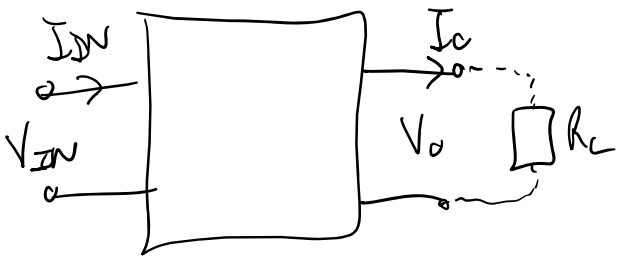
Lecture 3

Resistors are not the only passive element

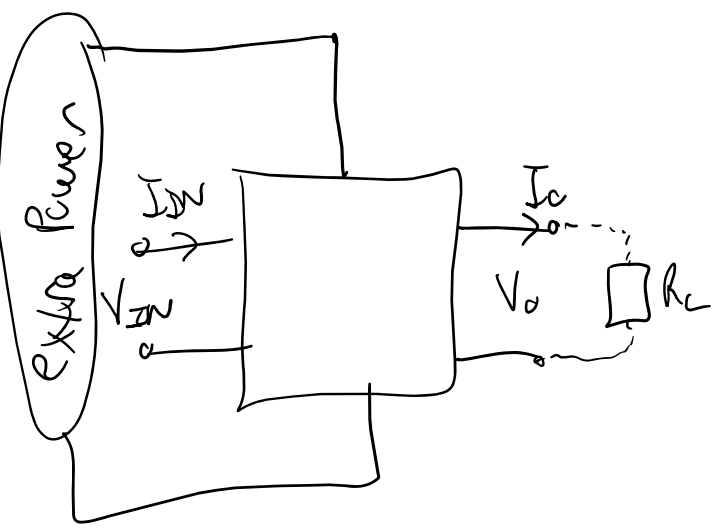
↓ No extra power

- transistors are "active" elements b/c they require additiant power
- operational amplifiers → active elements

Passive element



Active element



Other Passive elements.

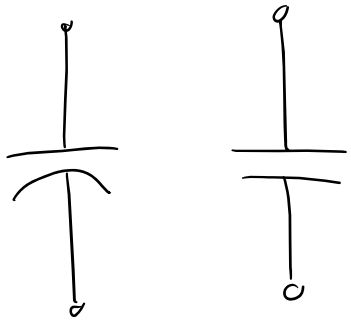
Capacitors

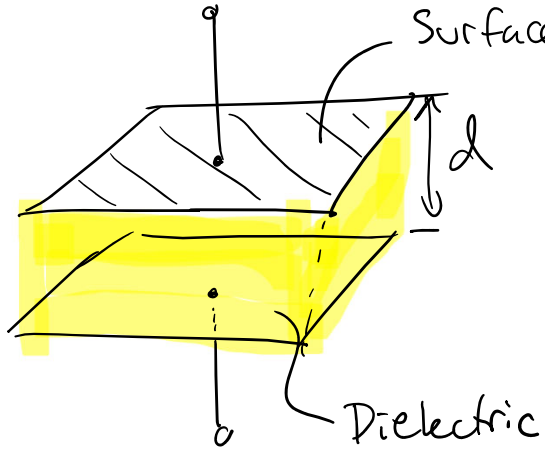
inductors

Diode

Resistors

Capacitors





* Capacitors have a unit of measure (Farad's) [F]

$\varepsilon_{air} \approx 8.85 \frac{pF}{m}$

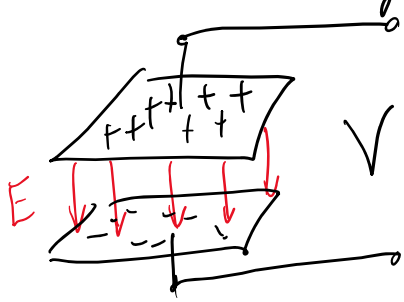
$\varepsilon_0 \rightarrow$ freespace

$\varepsilon_0 = 8.85 \frac{pF}{m}$

$$C = \varepsilon \frac{S}{d} \; [F]$$

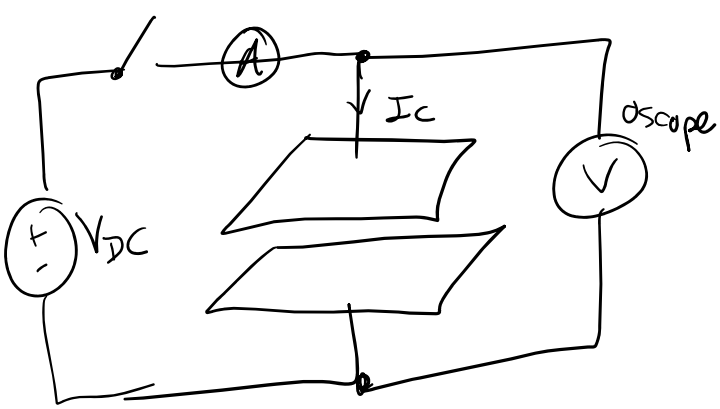
ε (permittivity)
electrical materail property

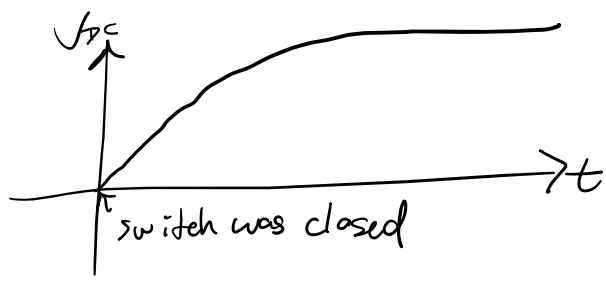
* capacitors store charges (Q) on the plates

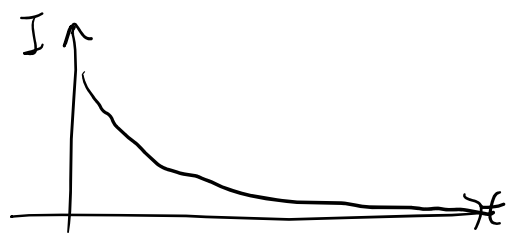


$Q = CV \rightarrow C = \frac{Q}{V}$

* Capacitor is an electrical spring.



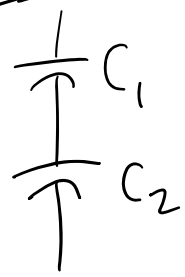




$$I_c = C \frac{dV_c}{dt}$$

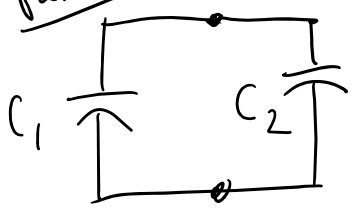
Aside (Battery)

Series



$$\Rightarrow \quad C_T = \frac{1}{\left(\frac{1}{C_1}\right)+\left(\frac{1}{C_2}\right)}$$

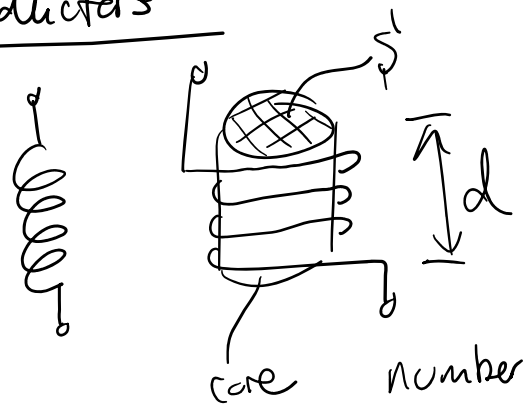
Parallel



$$\Rightarrow \quad C_T = C_1 + C_2$$

Inductors

Inductance $(L) \rightarrow$ measured in Henrys $[H]$



$$L = N^2 \mu \frac{S}{d}$$

μ = permeability $\left[\frac{H}{m}\right]$

number of turns (N)

in vacuum (μ_0)

$$\mu_0 = 4\pi \times 10^{-7} \left[\frac{H}{m}\right] = 1.26 \frac{\mu H}{m}$$

- Inductance is the ability to store magnetic fields
- Inductors store "current"

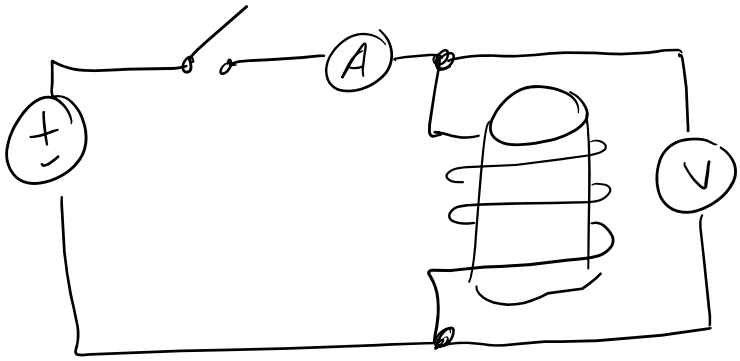
* Inductors are the electrical equivalent of a flywheel or gyroscope

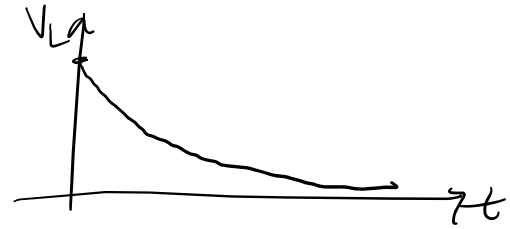
$$\Phi = LI \qquad P = m\vec{v}$$

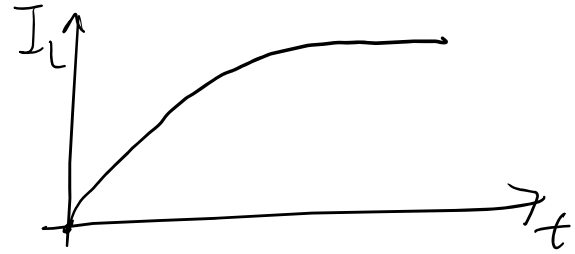
magnetic flux

momentum

$$L = \frac{\Phi}{I}$$



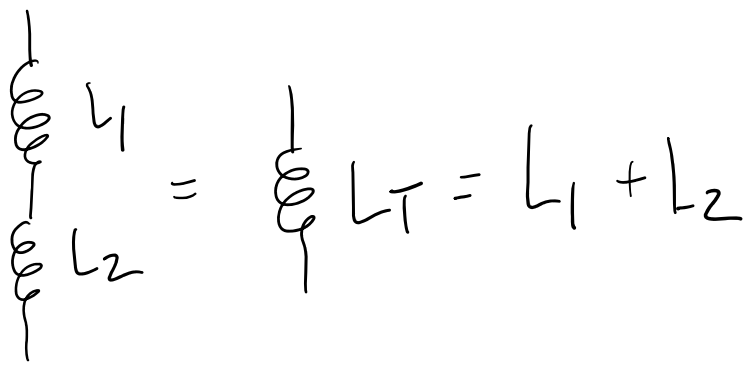


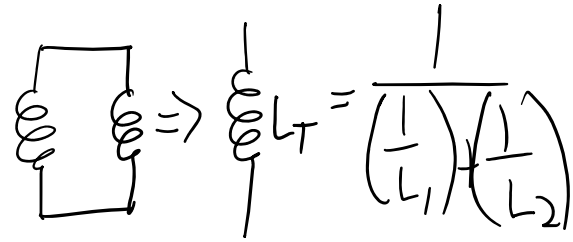


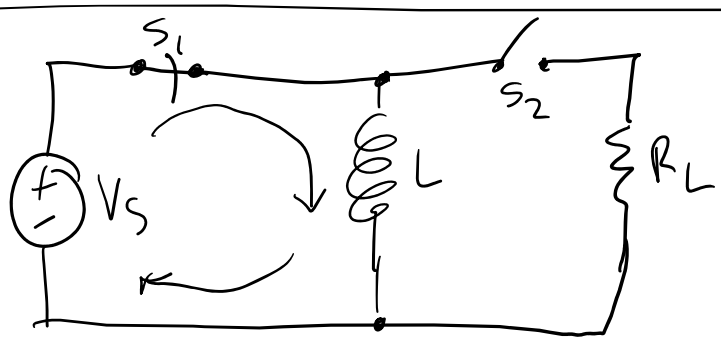
$$V_L = L\frac{dI}{dt}$$

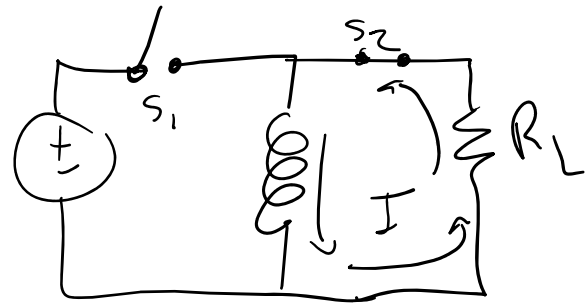
* Inductors are the "mass" in an electric circuit

<u>Inductors</u> (act just like resistors when series and parallel)







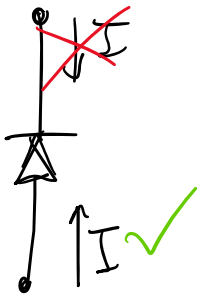


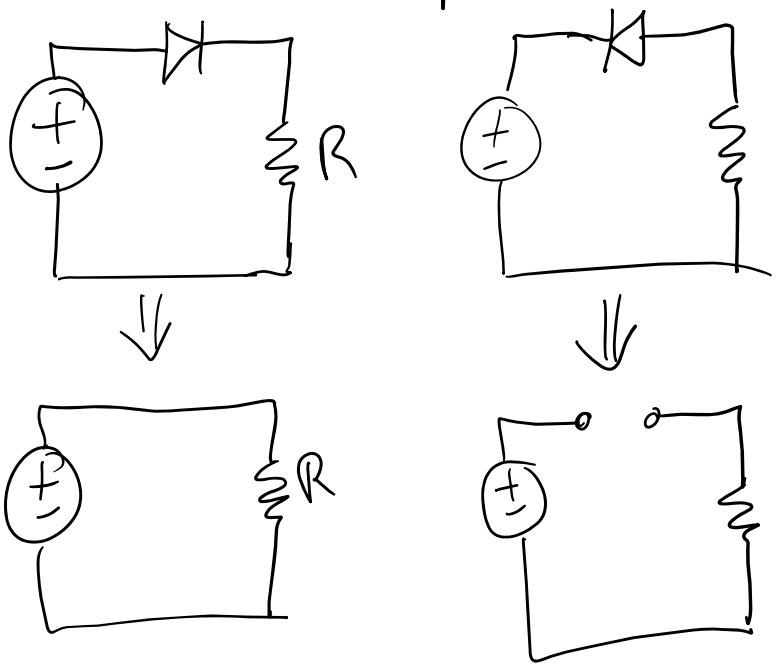
S_1 is closed and S_2 is open
- we "charge" the inductor.

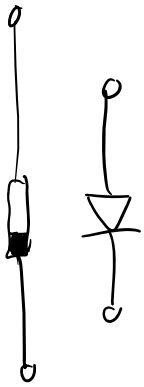
S_1 opens, S_2 closes
- inductor discharges through R_L

Diodes

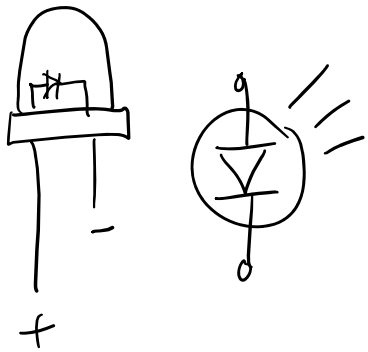
* current can only flow in one direction.

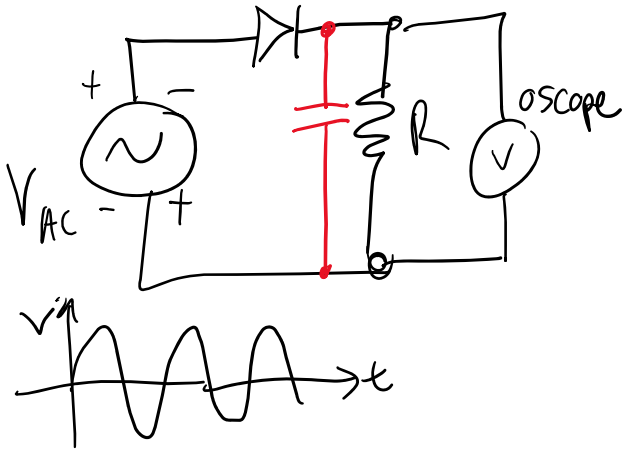


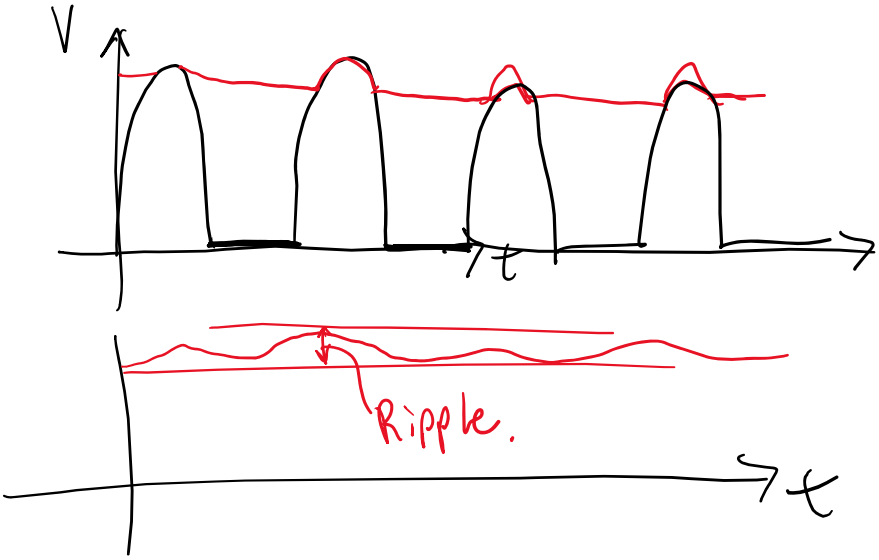




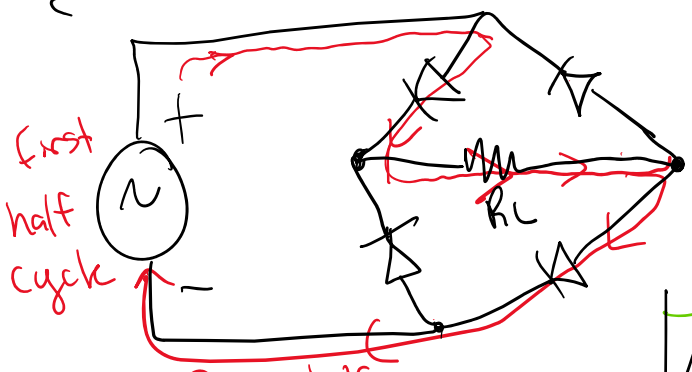
Light Emitting Diodes (LEDs)

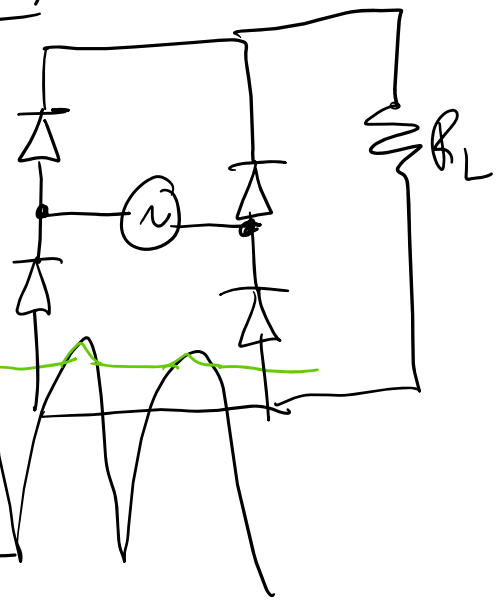


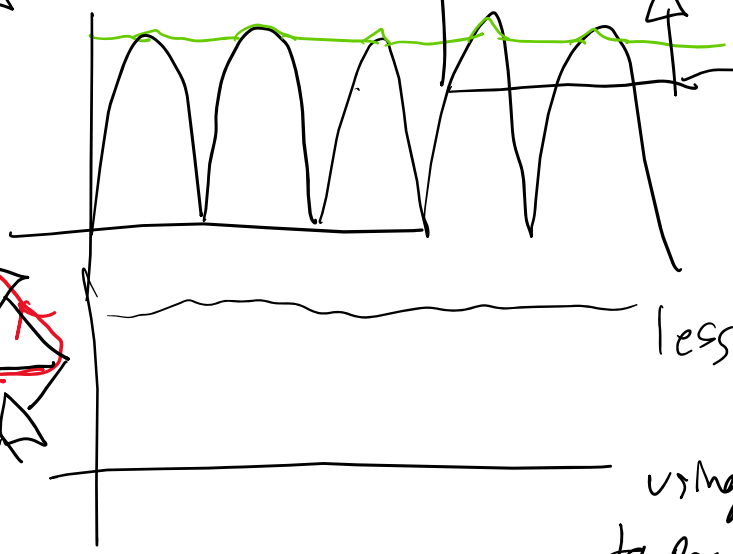




full-bridge Rectifier (full-wave)







less Ripple

using all cycles to power the Load